

Interface Trapping Effects on the Charge Transport Characteristics of LDPE/ZnO Nanocomposites

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ABSTRACT

This paper intends to reveal the interface trapping effect on the charge transport characteristics of LDPE/ZnO nanocomposite by combined study of the nanocomposite structures, trap level distributions, space charge distributions, conduction current and charge mobility. Experimental results show that nanofillers in the nanocomposites significantly altered the microstructure of the polymer matrix by introducing lots of smaller spherulites and increasing the crystallinity, leading to a significant increase in the interface areas between amorphous and crystalline regions. The trap density in the nanocomposites is about 2 - 6 times that in LDPE. The traps are essentially cavities, located at the interface areas between nanofillers and polymer matrix and interface areas between amorphous and crystalline regions. A new method was proposed to evaluate the carrier mobility as a function of discharge time and trap depth from the IDC results directly. The carrier mobility of LDPE is about 4-20 times higher than that of the nanocomposites. Compared with LDPE, bulk space charge accumulation is significantly suppressed in the nanocomposites. An interesting discovery is that there is a clear correspondence between the space charge accumulation and current increase in I-t characteristics under high electric field. The study indicates that the low charge carrier mobility and space charge suppression in LDPE/ZnO nanocomposites and the I-t behavior are all closely related to the interface trapping effects in the nanocomposites.

Index Terms - LDPE/ZnO, carrier transport, space charge, charge trap.

1 INTRODUCTION

POLYMER dielectric materials are apt to capture charge carriers to form space charges under high electrical field, which is an especially major restricted factor for HVDC (high voltage direct current) cable development [1]. Space charges induce kinds of detrimental effects, seriously distorting the electrical field, generating hot electrons, leading to electrical-mechanical energy storage and release, etc [2]. Thus, the investigation of charge transport characteristics and the understanding of space charge behavior and its suppression have become a main focus in the development of HVDC cable. However, despite long and widely study [3-7], there is still scant knowledge about the charge transport and space charge suppression in polymers. This is due to the complex structure of polymers as compared with crystals. In polymers, the interaction between equivalent electronic states in the same chain and the neighboring chain is weak and only narrow conduction bands are formed. So electrons are generally localized as a result. The localization is mainly

caused by chemical and structure inhomogeneities (impurities, chemical changes in the chain, chain ends, kinks, folds, etc.), which give rise to trapping states. The charge transport of polymers is largely dependent on the energetic position and occupancy of the trapping states. Yet it is still far from a complete understanding of the essence of traps and their effect on charge transport.

On the other hand, polymer dielectrics with improved dielectric properties are urgently required with the growing pursuit of high power density, high reliability under complex, long-term durability for electrical equipment and electronic devices working in harsh conditions, etc.^[8, 9]. Recently, polymer based nanocomposite dielectrics have drawn much attention due to their remarkably improved electrical properties [10, 11]. It has been reported recently that the addition of nanosized MgO or ZnO particles to LDPE (low density polyethylene) improved the breakdown strength and enhanced electrical treeing life, especially suppressed space charge accumulation and decreased the electrical conductivity of polyethylene (PE) [12-14]. However, the mechanism for the enhanced dielectric properties of the nanocomposite dielectrics is still unclear. The vast variety of experimental

observations is explained by several models, each of which explains, to a certain degree, a particular experiment, but often fails to give a satisfactory interpretation of other experiments [15, 16]. The investigation on charge transport and trap characteristics can provide a deep insight into the problem. This paper intends to extract important information about the charge transport characteristics and to reveal possible mechanisms of space charge suppression in LDPE/ZnO nanocomposite in the view of the essentially important effect of charge traps. A model is proposed for interpretation of space charge suppression in the nanocomposites.

2 EXPERIMENTS

2.1 MATERIALS

LDPE is provided by Daqing petrochemical Company marked as 18D. The LDPE/ZnO nanocomposites were prepared by melt blending method with 0.1, 0.5, 1, 3, 5, 7 weight percent of ZnO fillers, referred to as N0.1, N0.5, N1, N3, N5 and N7, respectively. The preparation procedure has been described in detail in previous publications by the authors^[17]. All tested samples were evaporated with Al electrodes on both sides with a diameter of 20 mm.

2.2 SCANNING ELECTRONIC MICROSCOPE IMAGING

The sample section surface, fractured in liquid nitrogen, was observed by scanning electron microscopy (SEM, Philips Quanta 200) operated at 20 kV in order to observe the dispersion of nanofillers.

2.3 ATOMIC FORCE MICROSCOPE IMAGING

Atomic force microscope (Shimadzu, Japan) was used to characterize the surface morphology of LDPE and the nanocomposites after corona ageing. Corona ageing was carried out under 3 kV ac voltage for 2 hours with a needle-plate electrode system ageing system. The specimens used for corona ageing were single-face evaporated with Al electrodes ($d=20$ mm). 2 mm air gap was kept between the other face of the specimen and the needle electrode.

2.4 X-RAY DIFFRACTION

X-ray diffraction (XRD) was utilized to determine the crystallinity and crystal cell size of LDPE and the nanocomposites. The type of the XRD analyzer used was D8 ADVANCE (Bruker2AXS Corporation, Germany). The scanning was taken between $10-80^\circ$ (2θ) with scanning speed of 8° (2θ) per min. The operating target voltage was 80 kV and the current was 10 mA. The radiation source was CuK alpha with wave length of $1.545 \text{ \AA}$.

2.5 SPACE CHARGE DISTRIBUTION MEASUREMENT

Space charge distribution measurement was performed based on pulsed electro-acoustic (PEA) method under dc 50 kV/mm at room temperature. The principle and application of PEA method has been widely discussed in previous publications^[18]. The instrument was marked as

FIV-PEANUTS provided by Japan Five Lab Co., Ltd. The space resolution and charge resolution of the instrument is about $10 \text{ }\mu\text{m}$ and 0.16 C/m^3 respectively.

2.6 ISOTHERMAL DISCHARGE CURRENT (IDC) MEASUREMENT

The specimen was firstly polarized at 323 K with a negative dc electric field of 30 kV/mm for 40 min. Then at a constant temperature 323 K, the discharge current generated by the detrapped electrons was measured immediately after the specimen was short-circuited for 20000 s. Only detrapped electrons are considered since the band diagram is normally favorable to electron injection [19].

2.7 CONDUCTION CURRENT MEASUREMENT

Conduction current was measured by a picoammeter EST122 and a data acquisition system. The negative dc electric fields of 10, 30 and 50 kV/mm were sequentially applied for 1 h under each electric field. The measurement was performed under room temperature in a drying oven.

3 METHODS

3.1 DETERMINATION OF TRAP LEVEL DISTRIBUTION BY MODIFIED ISOTHERMAL DISCHARGE CURRENT (MIDC) METHOD

This method has been presented in detail in the published papers by the authors [20]. Only the main results are shown below. In case of a continuous trap level distribution with the function, which is spatially uniform deep into the film with a distance l , the induced current in the external circuit during IDC measurement can be calculated as follows,

$$I(t) = \int_{E_v}^{E_c} \left[\int_0^l e f_0(E) N_t(E) S \frac{x}{d} e_n(E) e^{-e_n(E)t} dx \right] dE \quad (1)$$

$$= \frac{e S l^2}{2d} \int_{E_v}^{E_c} f_0(E) N_t(E) e_n(E) e^{-e_n(E)t} dE$$

where $N_t(E)$ is the trap energy level density function, e is the electronic charge quantity, $f_0(E)$ is the initial occupancy of a trap level and is a constant for a certain injection condition, S is the area of the electrodes, E is the trap level, v is the escape frequency of trapped electrons (10^{12} is advised by Toomer and Lewis [21]), l is the penetration depth of the injected electrons (It can be approximately evaluated from PEA measurement). $e_n(E)$ is the probability per unit time of an electron excited from traps with trap depth to the conduction band and can be expressed as [17]

$$e_n(E) = 1/\tau = v \exp(-E_t/kT) \quad (2)$$

where τ is the relaxation time, k is the Boltzmann constant. d is the thickness of the sample. By proper approximation and carrying out some calculations, the following equations can be obtained.

$$E_t = kT \ln(vt) \quad (3)$$

$$I(t)t = \frac{e S l^2 k T}{2d} f_0(E) N_t(E) \quad (4)$$

Assume all traps are filled, and then the trap level distribution can be derived directly from equation (3) and equation (4) by plotting $I(t)t$ against $\ln(vt)$.

3.2 DETERMINATION OF CARRIER MOBILITY FROM IDC

The carrier mobility can also be obtained from the IDC measurements. The method is generally similar to that provided by Montanari who obtained the carrier mobility from PEA measurement [22].

During the IDC test process, space charges are released and returned back to the injected electrodes under internal electric field produced by remaining space charges and thermal stimulation as time goes correspondingly, the current of IDC can be induced in the external circuit. So the carrier mobility as a function of time can be deduced from the current $I(t)$ and variations of residual space charges $q(t)$ as followings.

$$q(t) = \int_0^t I(t) dt \quad (5)$$

$I(t)$ can be expressed as,

$$I(t) = \mu(t)\rho(t)E(t)S = \mu(t)E(t)lq(t) \quad (6)$$

where μ is the carrier mobility, ρ is the space charge density, E is the Poisson electric field and can be expressed as l is the insulation thickness and S is the electrode area. By substitution of $q(t)$ and $E(t)$ in equation (6), we can draw the carrier mobility

$$\mu(t) = \frac{I(t)\epsilon l S}{(\int_0^t I(t) dt)^2} \quad (7)$$

Furthermore, we can obtain the carrier mobility as a function of trap depth E_t according to equation (3).

$$E_t = kT \ln(vt) \quad (8)$$

where k is the Boltzmann constant, T is temperature, v is the attempt-escape frequency (10^{12} s^{-1} is advised as showed in IDC methods), t is the time and approximates to the trapped charge carrier life time τ as showed by IDC methods.

4 RESULTS AND DISCUSSION

4.1 NANOPARTICLE DISPERSION CHARACTERIZATION

Dispersion of the ZnO nanoparticles in the nanocomposite N1 is shown by SEM image in Figure 1. The ZnO nanoparticles are uniformly distributed in LDPE. The particle size is in the range 40 - 100 nm, with most particle size around 50 nm. Few large agglomerates can be observed.

4.2 CRYSTALLINITY AND SPHERULITE SIZE

Figure 2 shows the XRD spectra of LDPE and the nanocomposite N5. There are two sharp diffraction peaks at $2\theta=21.43^\circ$ and $2\theta=23.87^\circ$, which is associated with the orthorhombic structure of LDPE and corresponds to the diffractions of 110 and 200 faces in crystalline phase. A broad shoulder located at 19.3° is observed next to the (110) peak caused by the amorphous phase. The crystalline peaks of

polyethylene can be separated from the amorphous area by the curve fitting as shown by Figure 3. The parameters such as full width at half maximum (FWHM) and crystallinity can be obtained from the crystalline peaks. The FWHM of crystalline peaks of polyethylene, i.e., FWHM (110) and FWHM (200), can be employed to determine the perfectness of polyethylene crystals in LDPE and LDPE/ZnO nanocomposites. The crystallinity of LDPE and N7% are 41% and 48% respectively, which are determined from the ratio of the peak areas of the crystalline and amorphous phases. The increase of crystallinity in N7% can be ascribed to the heterogeneous nucleation effect of the ZnO nanofillers as will be discussed in detail later on.

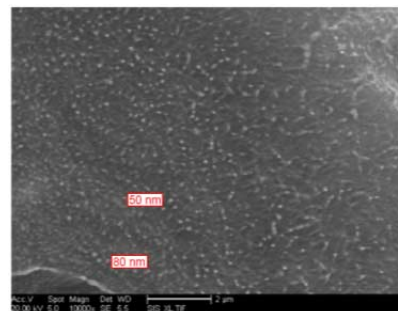


Figure 1. Sectional morphology graph of LDPE/ZnO 1%.

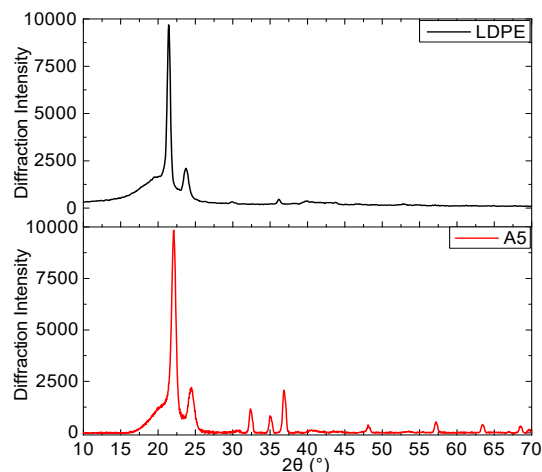


Figure 2. XRD spectra of LDPE and N1.

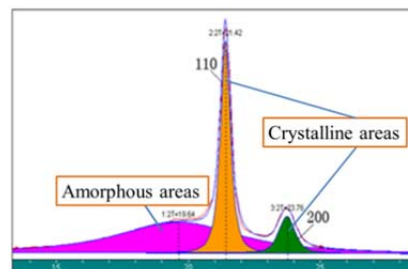


Figure 3. Peak separation of the XRD spectrum of LDPE.

Figure 4 shows the AFM morphology of LDPE and N1 after corona etching. Since the polyethylene spherulites shows much more strong corona resistance ability than the amorphous region, so the amorphous region on the surface of

the samples are etched and the left crystalline spherulites are left (bright areas) after the etching process. Figure 4 shows that the spherulite diameter of LDPE is about 1 to 2 μm , while the nanocomposite shows significantly reduced spherulite size with the diameter less than 0.5 μm but a significant increase in the spherulite number.

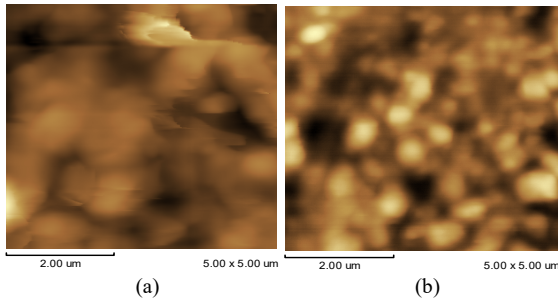


Figure 4 AFM morphology of corona etched LDPE (a) and 1wt% LDPE/ZnO(N1) nanocomposite (b).

The significantly increased smaller spherulites and crystallinity in the nanocomposite can be explained as follows. Since the nanofillers shows comparable size with polyethylene molecular chain length (10 - 100 nm), so it can act as heterogeneous nucleation agent and promote the formation speed of crystal nuclear during polyethylene crystallization process. On the other hand, a large number of nanofillers in the nanocomposite can significantly reduce the mobility of polymer molecules and increase the viscosity of the system due to steric hindrance effects. This can lower the spherulite growth rate. As a result of the heterogeneous nucleation and steric hindrance effects of nanofillers, a large number of small spherulites are formed uniformly in the nanocomposite, leading to smaller spherulite size and higher crystallinity in the nanocomposite. This explanation can be support by many other researches [23]. Huang et al studied the effect of SiO_2 nanofillers on the surface microstructure of LLDPE [24]. They found that there are obvious spherulitic structure with a spherulite diameter of about 10 μm formed by the folded lamella, while there are a large number of smaller spherulites in the nanocomposite.

4.3 SPACE CHARGE DISTRIBUTION

The space charge distributions in LDPE and LDPE/ZnO nanocomposites under different electric fields are showed in Figure 5. As for LDPE, notable heterocharges are formed around cathode electrode under 30 kV/mm. Under 50 kV/mm, both electrons and holes are separately injected symmetrically from the cathode and anode in LDPE immediately after the voltage is applied. After 900 s, a large number of heterocharges and homocharges are formed around the cathode, with a maximum density of about 6 C/m^3 . The injected electrons move toward anode with time.

For the nanocomposite N0.1, a large number of electrons and holes are symmetrically injected and the number of injected charges increases with an increase in electric field. Compared with LDPE, the injected charges located more close to the electrodes and move more slowly. For the nanocomposite

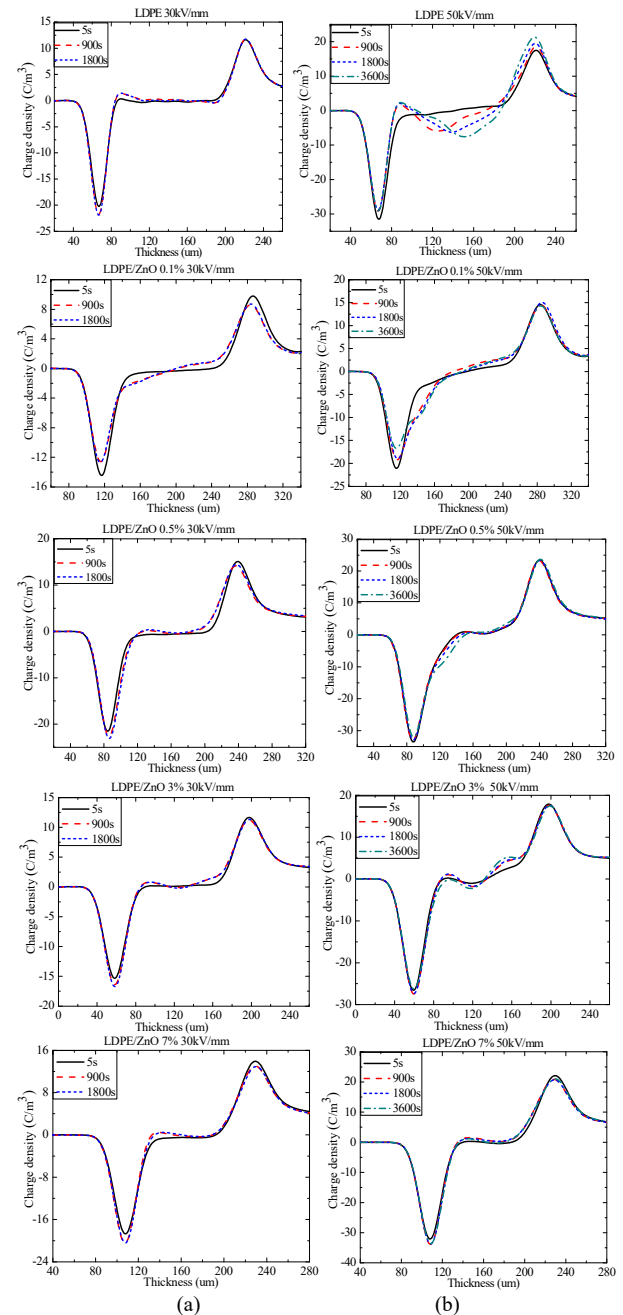


Figure 5 Space charge distributions in LDPE/ZnO nanocomposites under 30 kV/mm (a) and 50 kV/mm (b).

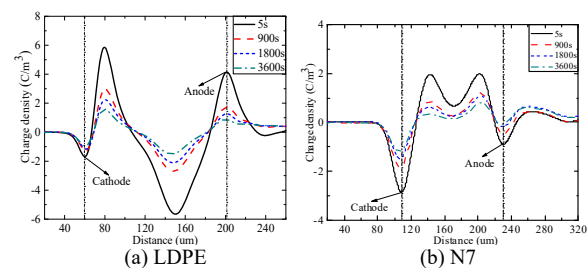


Figure 6 Space charge distributions in LDPE and N7 under short circuit conditions after polarized under 50 kV/mm for 1h.

N0.5, a small quantity of electrons and few holes are injected and homocharges are formed under both 30 kV/mm and 50

kV/mm. Similarly to N0.1, the injected electrons and holes are located more close to the electrode and largely immobilized compared to LDPE. No significant heterocharges are formed. Both the homocharges and heterocharges are largely decreased as compared with LDPE and N0.1. For the nanocomposite N3, a small number of heterocharges are gradually formed near the cathode. Much more holes are injected from the anode. For the nanocomposite N7, we are surprised to find that hardly any injected homecharges or ionized heterocharges are formed as compared with LDPE and other nanocomposites. It exhibits remarkable space charge suppression effect. The significant suppression of space charge in LDPE/ZnO nanocomposites is important for high voltage direct current cable development.

On the whole, space charge characteristics in the nanocomposites are significantly different from that in LDPE. Firstly, electron injection is dominant under high electric field in LDPE, while electron injection is relatively reduced and hole injection is relatively increased in the LDPE/ZnO nanocomposites. This can be ascribed to the fact that ZnO nanofillers promoted the formation of a large number of smaller spherulites in the crystallization process and increased crystallinity of LDPE as shown by XRD and AFM results. This is reasonable considering that the contact between crystalline region and Al electrode is ohmic for hole injection, while the contact between amorphous region and Al electrode is ohmic for electron injection as reported by other research [25, 26]. Secondly, the injected charges in the nanocomposite are much less, located more close to the electrode and moved more slowly than that in LDPE. Thirdly, heteropolar space charge was significantly suppressed in the nanocomposites. The reason should be that the increase in the crystallinity and introduction of lots of small spherulites can prevent the formation of large interphase macroscopic defects, which can restrict the ion mobility in a large range and weaken the ionization.

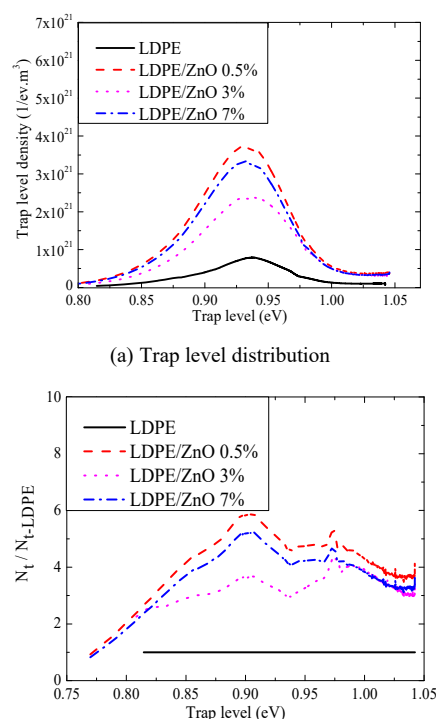
Figure 6 shows space charge distributions in LDPE and N7 under short circuit conditions after polarized under 50 kV/mm for 1h. The homopolar charges decay quickly immediately after short circuit, which indicates that these charges are from shallow traps (trap depth less than 0.9 eV). After about 900 s, the charges decay slowly and a few charges are remained in the deep traps even after 3600 s.

4.4 TRAP LEVEL DISTRIBUTIONS

Figures 7 shows the trap level distributions in LDPE and LDPE/ZnO nanocomposites with different filler contents. The results show that the trap levels are both in the range 0.8 - 1.05 eV with a peak around 0.94 eV, which can be regarded as deep trap levels in LDPE^[27]. This is consistent very well with the results obtained by M. Ieda, who assigned the deep trap level in LDPE to be 0.92 eV [6].

Trapping states in polymers are mainly due to chemical impurities and structure inhomogeneity. LDPE is a semi-crystalline polymeric material with most of the deep

trapping states mainly located in the interface areas, which has been demonstrated by Ieda via TSC and TL experiments [6]. In nanocomposite dielectrics, it has already been recognized that interface areas between the nanofillers and the polymer matrix play an important role in determine the electrical behavior of nanodielectrics [28]. Polymer molecules in the interface areas are partly immobilized due to their interaction with the nanoparticles and the space steric hindrance of the nanoparticles [29-30]. The change in the local structures (free volume, chain arrangement, mobility, etc.) of the interface polymers may increase the density and depth of the charge carrier trapping states [31]. The fact that nanofillers can lead to a change in the polymer morphology structure and an increase in the trap level density has been confirmed by numerous experimental results [31]. On the other hand, deep traps can also be formed at interphase areas between the crystalline and amorphous regions. As shown earlier by AFM and XRD results, nanofillers increased the crystallinity of LDPE and introduced a large number of small spherulites. This can increase the interface between crystalline region and amorphous region and then produce a large number of deep traps. Therefore, the nanofillers can significantly improve the trap level distributions. The properties, shape and dispersion of nanoparticles and compatibility with polymer have important influences on the structures of the nanocomposites and then on the trap characteristics.



(b) Ratio of the trap level density of the nanocomposites with LDPE

Figure 7. Trap level distributions of LDPE and LDPE/ZnO nanocomposites.

We will refer the two effects of nanofillers on trap characteristics of polymers as “interface trapping effect”, which can explain several electrical behaviors of polymer

based nanocomposite dielectric materials. The traps are further confirmed to be physically interface cavities located in the interphase regions between crystalline areas and amorphous areas or interphase regions between nanofillers and the polymer^[32].

The comparison of the trap level density is in the order of LDPE < N3 < N7 < N0.5. This may be due to the aggregation of nanoparticles is less and the dispersion is better in 0.5%, while the aggregation of the nanofillers in N3 and N7 is more serious, and the interface trapping effect caused by the nanofillers is also weakened. The trap density of N7 is greater than N3, which may be due to that the interface trapping effect of nanofillers is greater than the effect of agglomeration on the trap density with a further increase of the filler contents. The peak trap level density in LDPE and N0.5 are 8×10^{20} and $3.8 \times 10^{21} \text{ eV} \cdot \text{m}^{-3}$ respectively, which are of the same level as the existing results ($10^{16} - 10^{22} \text{ eV} \cdot \text{m}^{-3}$) [6, 19, 33, 34]. From Figure 7 (b), we can see that there is a significant increase in the deep trap density (0.88-1.0 eV) in the nanocomposites compared with LDPE.

4.5 CONDUCTION CURRENT

Figure 8 shows the conduction current of LDPE and LDPE/ZnO nanocomposites under 10 kV/mm, 30 kV/mm and 50 kV/mm. It can be clearly found that the conduction current in both LDPE and its nanocomposites decays with time. The current in LDPE decays more slowly and the steady current is higher than the nanocomposites. Considering the insignificant charge injection under 10 kV/mm, the current decay should be result from impurity ion polarization. The ion transport is more slowly due to the steric hindrance effect of nanofillers as discussed earlier, so the ion polarization is suppressed and the current is lower in the nanocomposite. Under 30 kV/mm and 50 kV/mm, the conduction current characteristics are obviously different from that under 10 kV/mm. Firstly, the current is about 100 - 1000 times higher than that under 10 kV/mm and current in the nanocomposites is much smaller (about 1/4 - 1/2) than that in LDPE. Secondly, the current in LDPE decays with time, while it shows a significant increase with time in N0.1 and N1. Moreover, the current in N0.5 and N7 remains nearly a steady value with time after it decays for the initial 300 s. Similar conduction current behaviors but with more strongly marked tendency are observed under 50 kV/mm. The current shows a remarkable increase in N0.1 and N1 and a weak increase in N0.5, while it exhibits a significant decrease in LDPE and keeps a steady value with time for N7. This is interesting considering that the conduction current is always found to show monotonous decay with time. Furthermore, it has been confirmed that the gradual increase, decay and changeless of the conduction current with time at high electric field are closely related to the accumulation of homopolar space charges, heteropolar space charges, and suppression of space charge formation, respectively^[35]. The abnormal characteristics of conduction current and space charge in the nanocomposite are dominated by the introduced abundant deep trapping states.

4.6 CARRIER MOBILITY

As can be seen from the Figures 9 and 10, the carrier mobility of LDPE and its nanocomposites decreases with

discharging time and the trap depth. In the time range 0 - 18000 s, the trap depth changes from about 0.8 - 1.05 eV, and the range of carrier mobility is $10^{-10} - 10^{-18} \text{ m}^2/\text{V} \cdot \text{s}$.

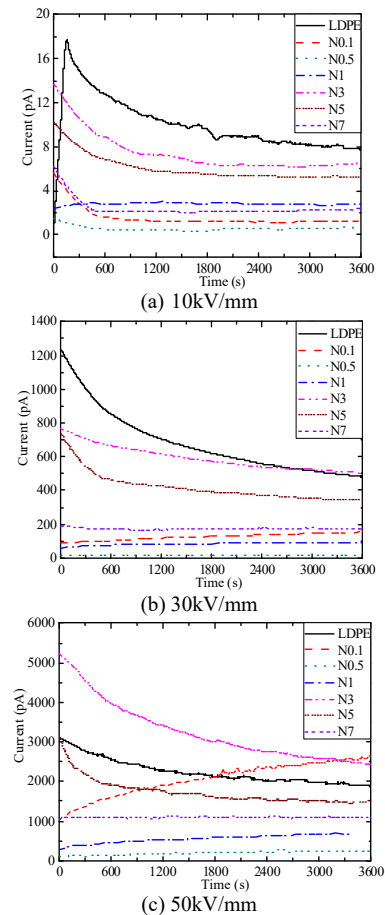


Figure 8 I-t characteristic of LDPE/ZnO nanocomposite under different electrical fields.

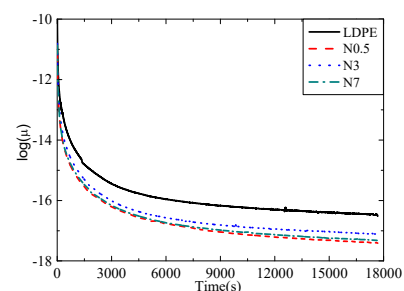


Figure 9 Carrier mobility variations with discharging time.

The range of mobility we obtained is similar to other results. J.M. Alison et al. reported that the mobility of XLPE obtained by isothermal discharge current method at room temperature in 0 - 10000 s is $10^{-14} - 10^{-16} \text{ m}^2/\text{V} \cdot \text{s}$ [36]. Z. Ziari reported that the mobility of LDPE under 20 kV/mm - 60 kV/mm at room temperature deduced from surface potential decay is also $10^{-14} - 10^{-16} \text{ m}^2/\text{V} \cdot \text{s}$ ^[37]. During discharging process, the charges in shallow traps are released earlier and transport with a higher effective mobility to the electrodes to induce a fast discharging current in the external circuit.

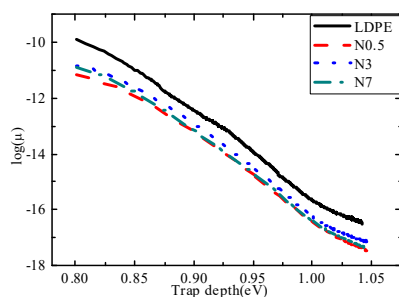


Figure 10 Carrier mobility variations with discharging time

It can be seen from Table 1 that the carrier mobility of nanocomposites is significantly less than LDPE, about 1/4 - 1/20 of LDPE, and the relationship between the size of carrier mobility to meet the LDPE > N3 > N7 > N0.5. It is well known that the trap depth and density have an important influence on the carrier mobility of the polymer, the deeper the trap level and the higher the density, the lower the carrier mobility. The trap level density of N0.5, N3 and N7 is about 5.3 times, 3.7 times and 5.0 times of that of LDPE, which is in full compliance with the carrier mobility changes.

Table 1 Carrier mobility of LDPE and its nanocomposites at different discharging time.

	1000s (m ² /V·s)	5000s (m ² /V·s)	10000s (m ² /V·s)	15000s (m ² /V·s)
LDPE	4.5×10^{-15}	1.4×10^{-16}	6.1×10^{-17}	4.0×10^{-17}
N0.5	6.9×10^{-16}	2.3×10^{-17}	7.8×10^{-18}	4.7×10^{-18}
N3	1.3×10^{-15}	3.6×10^{-17}	1.4×10^{-17}	9.3×10^{-18}
N7	8.1×10^{-16}	2.5×10^{-17}	9.2×10^{-18}	5.8×10^{-18}

5 CONCLUSIONS

The interface trapping effect on the charge transport characteristics of LDPE/ZnO nanocomposites has been revealed by combined study of the nanocomposite structures, trap level distributions, space charge distributions, conduction current and charge mobility. The following conclusions can be drawn.

- (1) Nanofillers in the nanocomposites significantly altered the microstructure of the polymer matrix by introducing lots of smaller spherulites and increasing the crystallinity due to the heterogeneous nucleation and steric hindrance effects of nanofillers, leading to a significant increase in the interface areas between amorphous and crystalline regions. In addition, large amounts of interface areas between nanofillers and polymer matrix are also introduced by nanofillers.
- (2) Both of the two kinds of interfaces introduced in the nanocomposites can form deep trap levels in the range 0.8 - 1.05 eV with a peak around 0.94 eV in the

nanocomposite. The deep traps are essentially cavities located at the interface areas. The interface trapping effect can explain several electrical behaviors of polymer based nanocomposite dielectric materials.

- (3) An interesting discovery is that there is a clear correspondence between the space charge accumulation and current increase in I-t characteristics under high electric field. The study indicates that the decrease of charge carrier mobility and space charge accumulation in LDPE/ZnO nanocomposites and the I-t behavior are all closely related to the significant increase of deep trap density.

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